

Soft Skills, Hard Times: Why New College Graduates Are Falling Behind

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Abstract

The unemployment rate for young college graduates has worsened in recent years. We document that this has happened along with a rise in demand for noncognitive skills by employers. We argue that these skills can only be accumulated on the job, so that the increase in demand for such skills harms young workers disproportionately, since they enter the labor market without any work experience. Using a search model calibrated to the US labor market, we show that skill biased technical change shifts employer demand towards high skilled workers, and this explains 36% of the worsening in the unemployment rate of young college graduates.

Keywords: noncognitive skills, youth unemployment, experience premium, skill demand

JEL classification: J23, J24, J64

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The purpose of college education is to prepare the youth for the labor force by providing them with the skills necessary to achieve a successful career. However, young college graduates in recent times are finding it increasingly difficult to get jobs relative to the average US worker. In this paper, we discuss why a college education in its current form is no longer enough to ensure success in the labor market.

Deming (2017), Lindqvist and Vestman (2011) and Edin et al. (2022) have documented the growing importance of soft skills in the labor market. We claim that these skills are primarily accumulated on the job. This implies that even if young college graduates have adequate technical skills, they may lack the soft skills needed by employers. As the demand for soft skills by employers increases, young college graduates are left at a disadvantage, experiencing worse labor market outcomes as a result.

Using the Current Population Survey (CPS) and data on the near-universe of online job postings in the US obtained from Lightcast, we document three empirical facts that motivate our interpretation. First, we show that historically, young college graduates were less likely to be unemployed than the average US worker.¹ In other words, a college education was associated with better labor market outcomes for workers in the past. However, we find that since 2011, the unemployment rate of young college graduates has been worsening, and since 2016, they are more likely to be unemployed than the average US worker.

Next, we show that coinciding with the rising unemployment rate of young college graduates, there has been an increase in the demand for soft skills by employers in the US labor market. For this exercise, we identify soft skills and technical skills in our job postings dataset using a combination of large language models and manual review, and we define skill demand as the average number of soft (technical) skills required in each job posting. Using this skill classification, we find that between 2011 and 2024, technical skill demand increased by 6.2%, possibly reflecting an environment where work is getting more specialized. However, during the same time period, soft skill demand increased by 15.6%.

Finally, we provide evidence that young workers fare worse in occupations that require more soft skills. Specifically, we show that the wage gap between young and older workers widens by 2.5-3 percent for each additional soft skill required in a job. This holds for both college educated workers and non-college educated workers. Interestingly, the relationship between technical skill demand and the wage gap depends on the education level of workers. For the non college educated sample, the wage gap between young and older workers also widens as technical skill demand increases. However, in the college educated sample, this wage gap becomes narrower in jobs that require more technical skills.

The wage gap patterns above suggest that although young college graduates have the relevant technical skills, they lack the soft skills required by employers. This can happen if soft skills are primarily accumulated on the job, which would imply that younger workers have had less opportunity to acquire them than older workers. Together, our three empirical facts indicate that an increase in the demand for soft skills can be an important factor behind the rising unemployment rate of young college graduates.

Using a general equilibrium search-and-matching model, we quantify how much of the rise

¹We define young workers as those aged 25 or younger. Older workers are between the ages of 26 and 55 years.

in the unemployment rate of young college graduates can be explained by the rise in demand for soft skills. Our model contains heterogeneous workers and firms, with on-the-job search and aggregate productivity shocks. Firms post vacancies, and both employed and unemployed workers receive job offers at random. Match creation is efficient, and wage determination happens through the sequential auctions framework developed by [Postel-Vinay and Robin \(2002\)](#). Workers can accumulate skills while employed. However, unemployment leads to skill depreciation.

Despite having heterogeneous workers and firms along with aggregate shocks, the model remains tractable. Worker flows and wage determination only depend on the match surplus, and the match surplus is independent of the current distributions of workers, vacancies, and matches ([Lise and Robin, 2017](#)). This means we can first solve for the surplus function, and then obtain the above distributions given the initial conditions. We calibrate this model to match key aggregate moments from the US labor market between 1951 and 2012.

In our counterfactual exercise, we incorporate skill biased technical change to generate an increase in skill demand by employers, equivalent to the rise in soft skill demand that we observe in the data over the 2011-2024 period. Due to skill biased technical change, all workers become productive, but high-skilled workers experience a larger increase in their productivity. This leads to a shift in the vacancy distribution towards higher skilled workers, which is the model equivalent of the rise in skill demand. Consequently, low skilled young workers face a lower share of viable vacancies, and their unemployment rate rises as a result. We show that this calibrated increase in skill demand generates approximately 36% of the observed increase in the unemployment rate of young college graduates.

Our paper contributes to the literature documenting the declining returns to cognitive skills and the rising importance of noncognitive skills. [Castex and Kogan Dechter \(2014\)](#) document that cognitive skills had a 30-50% larger effect on wages in the 1980s compared to the 2000s. [Beaudry et al. \(2016\)](#) and [Brynjolfsson and McAfee \(2014\)](#) attribute this to the rise in the adoption of IT, as computers can substitute for human workers by easily doing tasks requiring cognitive skills.

During the same time period, some papers show that noncognitive (or soft) skills have become an important determinant of labor market outcomes. Using Swedish military enlistment data, [Lindqvist and Vestman \(2011\)](#) observe that low levels of noncognitive skills are a strong predictor of failure in the labor market (i.e. unemployment), while high levels of cognitive (or technical) skills are necessary for success (i.e. high wages). Moreover, [Deming \(2017\)](#) in the case of the US, and [Edin et al. \(2022\)](#) in the context of Sweden show that the returns to noncognitive skills have increased over time. Our paper contributes to this literature by highlighting that noncognitive skills are typically accumulated on the job, making young workers the ones who suffer in the labor market.

Another set of papers discusses wage stagnation and the reduced demand for junior talent. [Guvenen et al. \(2022\)](#) document that median lifetime earnings in the US have stagnated. They show that this occurs primarily due to the stagnation of wages at the time of labor market entry. The [New York Federal Reserve](#) keeps track of the labor market outcomes of young workers, and it shows that wages have remained stagnant in recent times as well. [Bianchi and Paradisi \(2024\)](#) and [Citino et al. \(2026\)](#) document the difficulties faced by junior talent, which

has intensified in recent times due to remote work (Schubert, 2026; Emanuel et al., 2026).

Brynjolfsson et al. (2025) and Hosseini Maasoum and Lichtinger (2025) attribute the worsening outcomes of young college graduates to AI, while Emanuel et al. (2026) propose remote work as an explanation. AI can shift employer demand towards older workers, since AI may not have the context specific knowledge that comes with experience, although it can easily do intensive cognitive tasks that were previously assigned to young workers. Remote work can harm the productivity of young workers, as Emanuel et al. (2026) show that proximity to experienced workers is important for young, inexperienced workers to learn how their firm operates. Even though both AI and remote work are important determinants of young workers' outcomes in recent years, these reasons cannot explain why this worsening trend exists even in pre-COVID times. Our paper provides another mechanism at play: the rising demand for soft skills that are primarily accumulated on the job. This skill demand mechanism should be considered in conjunction with the AI and remote work mechanisms to explain the current difficulties faced by young workers.

The closest paper to ours is Wright (2026). They document a 50% decline in the employment rate of teenagers since the 1990s, and they attribute it to crowding out by adult workers whose own labor market opportunities have worsened. Although similar in nature, our paper focuses on young college graduates, and we attribute their worsening labor market outcomes to rising demand for soft skills.

The rest of the paper is organized as follows. First, we describe our data sources and the classification of soft and technical skills in Section I. We then document our empirical results in Section II. In Section III, we describe our model, and discuss its parametrization in Section IV. Section V contains our counterfactual exercise. We conclude in Section VI.

I Data

Our time period of interest is 2011-2025. We remove part-time workers, workers who are self-employed, and those working in the military. We classify workers as young if they are between 18-25 years old. Workers who are 26-55 years old are classified as old workers. A worker is defined to be college educated if she reports receiving a bachelor's degree or higher, otherwise she is considered as non-college-educated. For the wage analysis, we restrict our attention to the Monthly Outgoing Rotation Group (MORG). We use weekly earnings as our main wage variable, and convert it into real terms using 2025 dollars.

To measure labor demand in the US—specifically the demand for skills—we use online job postings data from Lightcast, formerly known as Burning Glass Technologies (Lightcast, 2024). Lightcast is a private business analytics company that collects data on online job postings from over 160,000 unique sources across the world each day. They use machine-learning algorithms on each posting to extract and infer several important details about each job posting such as its firm, location, occupation and industry classification, minimum experience required, and the exact date when the vacancy was posted. Most importantly for our purposes, we observe the specific skills that an employer demands in a job, at the job posting level.

The Lightcast dataset spans from January 1, 2011 until December 31, 2024. We remove job postings for internships, and we remove any job posting with missing posted date, occupation,

industry, job title, firm name, or skills demanded. This gives us approximately 229 million unique online job postings to analyze.

I.A Defining and Measuring Skill Demand

The Lightcast data contains 34,351 unique skills that are extracted from job postings. For our analysis, we want to classify these skills as either soft skills or technical skills (also known as noncognitive and cognitive skills, respectively). To identify soft skills mentioned in the job postings, we refer to the following definition given in the Occupational Information Network (O*NET, 2026):

“[Soft skills are] interpersonal and thinking skills needed to interact successfully with people and to perform efficiently and effectively in the workplace.”

We create our classification of soft and technical skills as follows: First, we sort all 34,351 skills based on the share of job postings they are mentioned in. Then, for the first 1,000 skills (which are present in $> 0.2\%$ of all job postings), we manually identify soft skills based on the ONET definition given above. For the remaining 33,351 skills, we feed them through a large language model (LLM) and ask it to identify soft skills based on the ONET soft skills definition.² All the skills that are not categorized as soft skills are classified as technical skills instead. Table 1 shows the most common soft and technical skills found in our dataset.

To measure skill demand, we propose the following measurement. Our measure simply calculates the average number of skills mentioned per job posting. We calculate this separately for soft skills, technical skills, and all skills.³

II Empirical Analysis

In this section, we will provide empirical evidence in support of our claim that rising demand for social skills is an important driver for the worsening labor market outcomes of young college graduates. First, we describe what is happening to the unemployment rate of young college graduates.

Figure 1 shows the unemployment rate of young college graduates in the US over time, along with the overall unemployment rate for reference. Between 1981 and 2011, young college graduates consistently experienced a relatively low unemployment rate, indicating that they are attractive to employers and find it easier to find jobs compared to the rest of the population. However, since 2011, the gap between these 2 lines has been shrinking, and since 2015, the youth unemployment rate has been higher than the overall unemployment rate.

²Our soft skill classification remains unchanged if we skip the manual identification step, and feed all 34,351 skills to the LLM. We obtain the ONET definition for soft skills <https://www.onetonline.org/skills/soft/>.

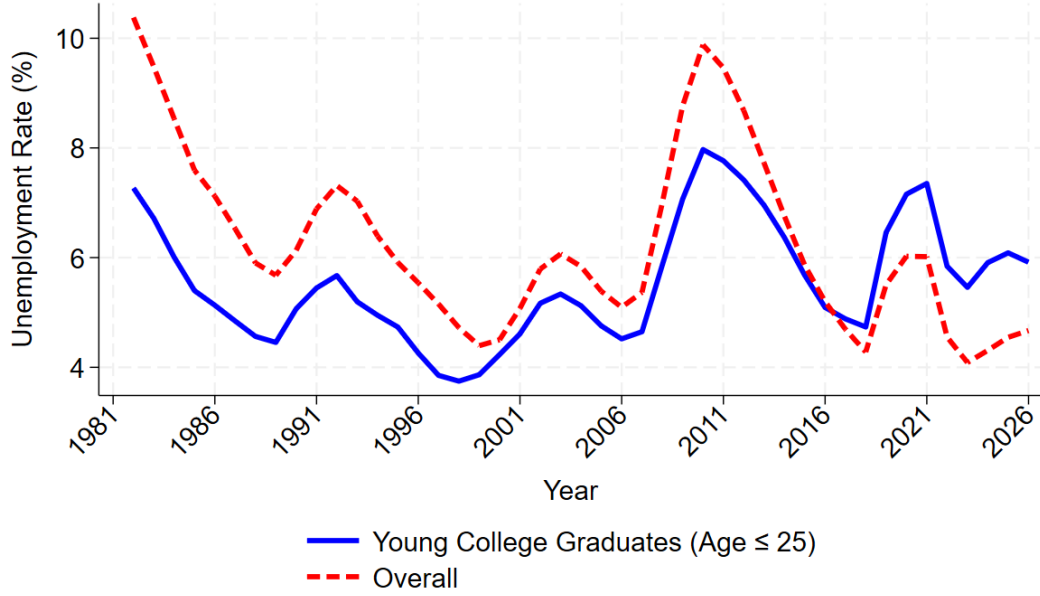
³A drawback of our skill demand measure is that it gives equal importance to all skills mentioned in a job posting. In Appendix A, we explore whether O*NET data on skill requirements can help us address this issue. We show that although O*NET data captures the relative importance of skills across occupations, it does not capture the aggregate growth in skill requirements over time that we can observe in Lightcast. Then we discuss how we utilize AI to address these limitations by using O*NET data to classify the soft skill and technical skill intensity of job postings in our Lightcast dataset.

Table 1: Most Common Soft and Technical Skills in Lightcast Job Postings, 2011–2024

Panel A: Soft Skills					
#	Skill	Share (%)	#	Skill	Share (%)
1	Communication	33.76	16	Time Management	5.72
2	Customer Service	28.97	17	Presentations	5.64
3	Management	25.59	18	Troubleshooting (Problem Solving)	5.55
4	Sales	19.70	19	Professionalism	5.29
5	Operations	16.93	20	Selling Techniques	5.12
6	Leadership	15.63	21	Teamwork	4.99
7	Problem Solving	11.51	22	Prioritization	4.46
8	Detail Oriented	11.48	23	Coaching	4.44
9	Planning	10.16	24	Decision Making	3.98
10	Interpersonal Communications	8.18	25	Cleanliness	3.95
11	Coordinating	6.96	26	Innovation	3.87
12	Multitasking	6.83	27	Accountability	3.68
13	Verbal Communication Skills	6.78	28	Influencing Skills	3.23
14	Organizational Skills	6.71	29	Analytical Skills	3.12
15	Self-Motivation	6.33	30	Mentorship	2.99
Panel B: Technical Skills					
#	Skill	Share (%)	#	Skill	Share (%)
1	Writing	10.87	16	Microsoft PowerPoint	4.32
2	Microsoft Excel	8.48	17	Mathematics	4.25
3	Valid Driver's License	8.33	18	Warehousing	4.05
4	Registered Nurse (RN)	8.15	19	Basic Life Support (BLS) Certification	4.04
5	Project Management	7.66	20	Microsoft Outlook	4.03
6	Merchandising	7.43	21	Computer Science	4.02
7	Microsoft Office	7.41	22	Restaurant Operation	3.77
8	Research	7.01	23	Workflow Management	3.04
9	Marketing	6.76	24	SQL (Programming Language)	2.98
10	Computer Literacy	6.62	25	Microsoft Word	2.89
11	Nursing	6.61	26	Product Knowledge	2.82
12	Accounting	6.08	27	Process Improvement	2.80
13	Auditing	5.85	28	Billing	2.80
14	Scheduling	5.77	29	Invoicing	2.74
15	Finance	4.98	30	Cash Register	2.71

Notes: This table contains the 30 most common soft and technical skills present in Lightcast job postings data. Lightcast identifies 34,351 unique skills in its job postings. Soft and technical skills are identified by a combination of large language models and manual review. The share column represents the percentage of job postings between 2011 and 2024 that ask for a particular skill.

Figure 1: The Unemployment Rate Of Young College Graduates Is Rising



Notes: This figure shows the time series for the unemployment rate of young college graduates, and for the overall unemployment rate in the US (Current Population Survey, 1982-2025). A worker is considered young if she is aged 25 or lower. College graduates are defined as those workers who report holding a bachelor's degree or higher.

This reversal implies that there is something happening that impacts young college graduates disproportionately. Given that this trend starts from 2011, it cannot be explained by the AI boom or remote work alone, which have only been prevalent in the post-COVID period.⁴

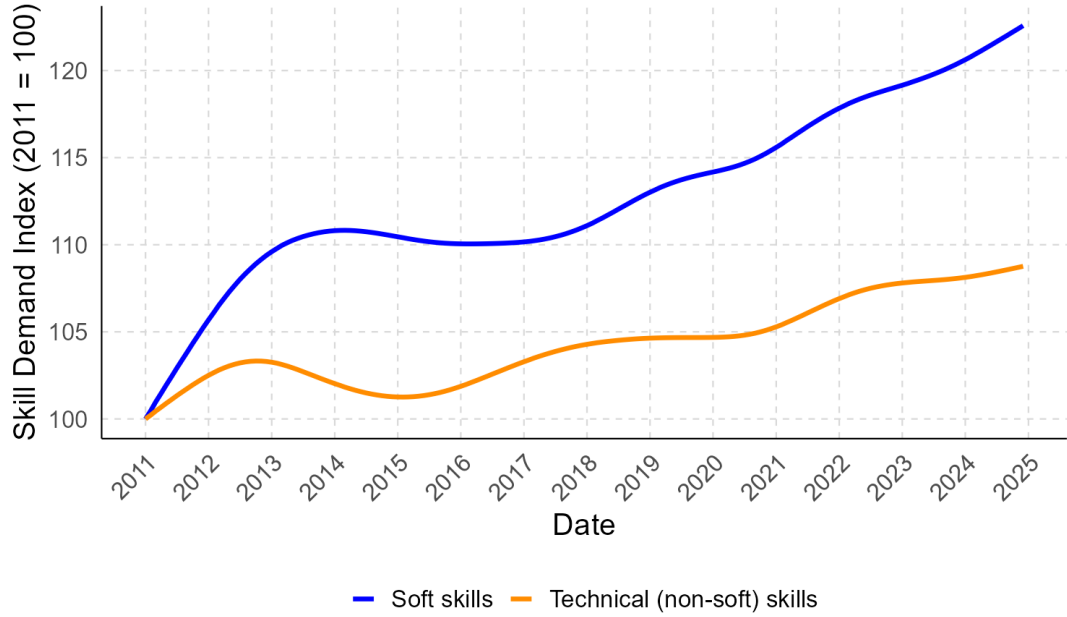
Using Lightcast data, we will show that the skill demand (specifically the demand for soft skills) has been increasing during the same period of time, and that it is an important driver behind this rise in the youth unemployment rate. Figure 2 plots the demand for soft skills and technical skills over time, and it shows that the demand for both skills has been increasing.

Figure 2 only shows the overall trend in skill demand. In Table 2, we regress skill demand on a linear time trend, after collapsing our dataset to the occupation $\times$ time level. We also control for occupation fixed effects to extract the average within-occupation increase in skill demand, and we control for month fixed effects to account for seasonality. To remove any effects from the increasing length of job postings, we also control for the average word count of a job posting within an occupation $\times$ time group. We find an annual increase of 1.2% in soft skill demand, and 0.48% in technical skill demand. This sums up to an overall increase of 15.6% in soft skill demand and 6.2% in technical skill demand over the period 2011-2024.

Next, we combine CPS and Lightcast data to show that young workers perform relatively better in occupations that require more technical skills, but they are worse off if their job requires higher social skills. First, using Lightcast data, we calculate the average number of

⁴In Appendix B, we show that over the same time period, the unemployment rate of young non-college graduates has fallen. If this group is able to gain more work experience by the time they reach the age of 25, this can explain why the labor market outcomes of young non-college graduates have improved, but the outcomes of young college graduates have not.

Figure 2: Time Series For Skill Demand



Notes: This figure show the growth in skill demand between 2011 and 2024 (Lightcast, 2011-2024). Soft and technical skills are identified by a combination of large language models and manual review. Soft (technical) skill demand is calculated as the average number of soft (technical) skills required in a job posting in a given month. The soft (technical) skill demand in January 2011 is normalized to 100. The time series is smoothed using an HP filter with $\lambda = 14,400$.

Table 2: Rising Demand For Soft Skills

Dependent Variable:	Log(All Skills)	Log(Soft Skills)	Log(Technical Skills)
% change per year	0.60*** (0.12)	1.20*** (0.24)	0.48*** (0.12)
<i>Fixed-Effects:</i>			
Occupation	Yes	Yes	Yes
Month	Yes	Yes	Yes

Notes: This table shows the slope of a linear time trend for skill demand after controlling for occupation and month fixed effects (Lightcast, 2011-2024). The unit of observation is an occupation $\times$ month $\times$ year group, where occupations are classified at the 6 digit SOC code level. Soft (technical) skill demand is calculated as the average number of soft (technical) skills required in a job posting in a given occupation $\times$ month $\times$ year group. All Skills = Soft Skills + Technical Skills. Coefficients are multiplied by 12 to reflect annual changes in skill demand. We also control for the average number of words per job posting within an occupation $\times$ month $\times$ year group. Each observation is weighted by the number of job postings within the group. Standard errors are clustered at the occupation level. $* \Rightarrow p < 0.1, ** \Rightarrow p < 0.05, *** \Rightarrow p < 0.01$.

soft skills and technical skills required per posting in each occupation $\times$ time group. Then we merge this data with CPS data, so that we can replace occupation fixed effects by the bundle of skills required in those occupations instead. Then we regress wages obtained by a respondent in their job in CPS on the levels of skills required in that job. Most importantly, we interact the skill demand measures with an indicator for young workers, to understand whether young workers are differentially affected in jobs requiring higher skill levels. Formally, we estimate

$$\begin{aligned}\log w_{iot} = & \beta_s \text{Soft}_{ot} + \beta_\tau \text{Tech}_{ot} + \gamma \text{Young}_i \\ & + \delta_s (\text{Young}_i \times \text{Soft}_{ot}) + \delta_\tau (\text{Young}_i \times \text{Tech}_{ot}) \\ & + X_i' \Theta + \lambda_t + \varepsilon_{iot},\end{aligned}$$

where i indexes CPS respondents, o occupations, and t months; Soft_{ot} and Tech_{ot} are the average soft and technical skills per posting in occupation o in month t , taken from Lightcast; Young_i is an indicator for whether respondent i is 25 years old or younger; X_i contains sex, race, and education; and λ_t are month fixed effects. The coefficients of interest are δ_s and δ_τ , which capture whether young workers are paid differentially in occupations requiring more soft or technical skills. We estimate this specification separately for college-educated and non-college-educated workers, using CPS MORG wage data from 2011–2024 for wage-employed respondents aged 55 or younger, merged with Lightcast skill demand at the occupation $\times$ month level.

Table 3 shows the results from this regression exercise. First, the coefficient on *Young* is large and negative, indicating that young workers earn less than old workers on average. Second, the coefficients on both soft skills and technical skills are positive, implying that workers in more skill intensive jobs will earn more on average. Importantly, the regression results show that not only technical skills, but also soft skills are valued in the labor market, as signified by the positive and significant coefficient assigned to it.

Finally, we focus on the interaction terms between *Young* and the skill demand measures. We find that the coefficient on $\text{Young} \times \text{Soft}$ is negative, both for the college educated and non-college educated sample. This means that young workers are worse off in jobs that require higher levels of soft skills. On the other hand, the coefficient for $\text{Young} \times \text{Technical}$ is positive for college educated workers but negative for non-college educated workers.

These results are consistent with an environment where college educated workers are prepared for the level of technical skills that are asked for in the labor market. However, even college students do not fare well in soft skill intensive jobs, indicating that soft skills seem to be accumulated on the job.

In **Appendix B**, we try different variations of our baseline specification. In our first specification, we recalculate the soft and technical skill demand at the occupation level instead of the occupation $\times$ month level, to keep the skill requirements of an occupation fixed over time. In our second specification, we include occupation fixed effects in our baseline regression, so that we can exclude any variation across occupations and focus on within occupation variation instead. In our final specification, we control for age and age squared, to remove any variation in wages that could be explained by work experience. **Table B1** and **Table B2** show that our results remain robust across all the above mentioned specifications.

Table 3: Young Workers Earn Less Than Older Workers

Dependent Variable: Log(Wages)	College Educated	Non-College Educated
Young (γ)	-37.24*** (3.96)	-21.85*** (2.50)
Soft skills (β_s)	2.66* (1.55)	0.72 (1.13)
Technical skills (β_t)	2.96*** (0.31)	3.55*** (0.45)
Young $\times$ Soft (δ_s)	-3.00** (1.45)	-2.52*** (0.95)
Young $\times$ Technical (δ_t)	0.93*** (0.33)	-0.80*** (0.27)
Observations	572,091	931,070

Notes: This table shows the differential returns from soft and technical skills to young and older workers (CPS Merged Outgoing Rotation Group + Lightcast, 2011 - 2024). The wage variable reports hourly wages. For CPS respondents for whom this data is missing, hourly wages are calculated by dividing weekly wages by usual hours worked per week (if this data is available). Young = 1 if age ≤ 25 , and 0 if age $\in [26, 55]$. A respondent is college educated if she reports obtaining a bachelor's or advanced degree. The Soft (Technical) skills variable is the average number of soft (technical) skills required per job posting in an occupation $\times$ month group in Lightcast. The regression controls for standard demographic variables and time fixed effects. Standard errors are clustered at the occupation level. * $\Rightarrow p < 0.1$, ** $\Rightarrow p < 0.05$, *** $\Rightarrow p < 0.01$.

In [Figure B3](#), we analyze whether the returns to soft and technical skills has changed over our sample period. We find that these returns remain remarkably consistent over time (although the returns to technical skills seem to be on a slightly downward trajectory). This implies that the change in wages due to a unit increase in skill requirements for an occupation remains the same across time. This evidence - combined with the observation that the demand for skills is increasing - would indicate that the gap between the wages of young and old workers will rise as a result, making young workers worse off.

Our evidence suggests why rising soft skill demand may worsen the youth unemployment rate. Young workers typically perform worse in jobs that require more soft skills, since these skills can only be learnt on the job, so young workers enter the labor market with low levels of these skills. Over the last few years, the demand for soft skills has increased, making jobs more soft skill intensive, and worsening the labor market outcomes of young college graduates. Since soft skills are accumulated with experience, employers then shift their demand towards more experienced workers, and young workers face a higher unemployment rate as a result.

III Model

We will now build a general equilibrium, search and matching model of the US labor market, using the model developed by [Lise and Robin \(2017\)](#) as a baseline. This model will allow for heterogeneous worker and firm types. The goal of the model is to quantify how much of the worsening labor market outcomes of young college graduates can be attributed to the rise in demand for social skills by employers.

III.A Workers, Firms, and States

Time is discrete. The economy contains a continuum of infinitely-lived workers identified by their skill level $x \in \mathcal{X} = [0, 1]$, and a continuum of firms characterized by productivity $y \in \mathcal{Y} = [0, 1]$. A worker's skill level is fixed over her lifetime, and the distribution of worker skills $\ell(x)$ is exogenous. The distribution of productivities across firms is uniform. Both workers and firms are risk neutral.

The aggregate state of the economy is given by z_t . At the beginning of each period t , the aggregate state evolves according to a Markov transition kernel $\Pi_z(z | z)$.

III.B Production and Flow Value of Unemployment

The output produced by a match between a worker with skills x and a firm with productivity y at date t is given by a second-order Taylor polynomial in the worker and firm type, scaled up by the level of aggregate shock z_t .

$$p_t(x, y, z) = z_t(a_1 + a_2x + a_3y + a_4x^2 + a_5y^2 + a_6xy) \quad (1)$$

This quadratic specification allows for the output to be non-monotonic in either the firm's or the worker's type. For example, if $a_5 < 0$, then it is possible that for some worker types x , moving to a high-type firm $y' > y$ leads to a lower level of output $p_t(x, y', z) < p_t(x, y, z)$. Similarly, the production function allows for complementarity between the worker and firm types (if $a_6 \neq 0$). Therefore, this specification will be an important driver of sorting between firms and workers, which will become apparent later.

For an unmatched worker, the flow value of unemployment is given as a fixed proportion of the output she can produce at her best possible firm, in the absence of any aggregate shocks:

$$b_t(x) = 0.7 \times p(x, y^*(x, 1), 1) \quad (2)$$

where the 0.7 factor is taken from [Hall and Milgrom \(2008\)](#), and $y^*(x, 1) = \arg \max_y p_t(x, y, 1)$ is the firm type that maximizes a worker's output in the absence of aggregate shocks ($z = 1$).

III.C Labor Markets

Consider the labor market at the beginning of period t . Let $u_t(x)$ denote the mass of unemployed workers with skills x , and let $h_t(x, y)$ denote the mass of employed workers with skills x currently matched with firm y . The stock of employed and unemployed workers should satisfy

$$u_t(x) + \int h_t(x, y) dy = \ell(x).$$

The timeline of the labor market is as follows: The period t starts with a change in the aggregate state from z_{t-1} to z_t . Separations between workers and firms happen, since some matches that were viable in period $t - 1$ (i.e., generated a positive surplus) are not viable anymore. The resulting stock of unemployed agents is denoted by $u_t^+(x)$ and the remaining employed workers are denoted by $h_t^+(x, y)$. The effective labor force (the mass of workers searching for jobs) is given by

$$L_t = \int u_t^+(x)dx + s \int \int_y h_t^+(x, y) dx dy, \quad (3)$$

where $s \in [0, 1]$ is the relative search efficiency of employed workers (the search effort of unemployed workers is normalized to 1).

During the search phase, firms of type y post vacancies $v_t(y)$. The total number of vacancies in the labor market is given by $V_t = \int v_t(y) dy$. Matches form according to

$$\mathcal{M}_t(L, V) = \min\{\alpha_M L^{1/2} V^{1/2}, V_t, L_t\}, \quad (4)$$

and market tightness is defined as $\theta_t = V_t/L_t$. The job finding rate is

$$\lambda_{w,t} = \frac{\mathcal{M}(L_t, V_t)}{L_t} = \alpha_M \theta_t^{1/2}, \quad (5)$$

and the job filling rate is

$$\lambda_{f,t} = \frac{\mathcal{M}(L_t, V_t)}{V_t} = \alpha_M \theta_t^{-1/2}. \quad (6)$$

Search is random. Each period, an unemployed worker receives a job offer with probability $\lambda_{w,t}$ and an employed worker receives a job offer with probability $s\lambda_{w,t}$. Similarly, a firm posting a vacancy gets in contact with a worker with probability $\lambda_{f,t}$.

III.D Values and Surplus

III.D.1 Unemployment

Let $B_t(x)$ denote the value of unemployment for a worker with skills x . If a worker remains unemployed throughout period t , she receives the value $b_t(x)$ (as defined in Equation 2). In period $t+1$, this worker can receive a job offer with probability $\lambda_{w,t+1}$. We assume that the worker has zero bargaining power, so any firm y matched with this worker will offer her reservation value $B_{t+1}(x)$. If the worker does not receive a job offer (with probability $1 - \lambda_{w,t+1}$), she remains unemployed and receives her value of unemployment $B_{t+1}(x)$. Therefore, the expression for $B_t(x)$ is given as follows:

$$B_t(x) = b_t(x) + \beta \mathbb{E}_t \left[(1 - \lambda_{w,t+1}) B_{t+1}(x) + \lambda_{w,t+1} \int \frac{v_{t+1}(y)}{V_{t+1}} B_{t+1}(x) dy \right] \quad (7)$$

$$= b_t(x) + \beta \mathbb{E}_t B_{t+1}(x) \quad (8)$$

where β is the discount factor and $\mathbb{E}_t$ is the expectation taken at time t with respect to the future aggregate states z .

III.D.2 Surplus

Let $P_t(x, y, z)$ denote the joint value of a match between a worker x and a type- y firm. Define match surplus as

$$S_t(x, y, z) = P_t(x, y, z) - B_t(x). \quad (9)$$

A match between a worker x and firm y produces output $p_t(x, y, z)$ in period t . In period $t + 1$, after the realization of the aggregate shock z_{t+1} , the worker and firm assess whether to continue with the match or not. Since matches are efficient, a match is destroyed if $P_{t+1}(x, y, z) < B_{t+1}(x) \Leftrightarrow S_{t+1}(x, y, z) < 0$ ⁵, otherwise the match remains viable (i.e., if $P_{t+1}(x, y, z) \geq B_{t+1}(x) \Leftrightarrow S_{t+1}(x, y, z) \geq 0$). Matches can also be destroyed with an exogenous probability δ in each period.⁶

If a match remains viable in period $t + 1$, the worker draws a job offer with probability $s\lambda_{w,t}$, and the probability that this job offer comes from a firm of type y' is $v_{t+1}(y')/V_{t+1}$. Whether the worker accepts or rejects the offer depends on the value offered by its current firm $W_{t+1}(x, y, z)$ and the competing firm $W_{t+1}(x, y', z)$. We use the sequential auction framework of Postel-Vinay and Robin (2002) to model the competition between firms and determine the wages offered to the worker.

Suppose the worker's current wages at firm y are $W_t(x, y, z) \leq P_t(x, y, z)$. If $P_{t+1}(x, y', z) > P_{t+1}(x, y, z) \Leftrightarrow S_{t+1}(x, y', z) > S_{t+1}(x, y, z)$, then the two firms compete with each other by raising their wage offers, until the wage offer reaches $W_{t+1}(x, y', z) = W_{t+1}(x, y, z) = P_{t+1}(x, y, z)$ and the worker switches to the new firm y' .⁷ On the other hand, if $P_{t+1}(x, y', z) \leq P_{t+1}(x, y, z) \Leftrightarrow S_{t+1}(x, y', z) \leq S_{t+1}(x, y, z)$, then the worker stays at its current firm, but it will receive a new wage $W_{t+1}(x, y, z) = \max[W_t(x, y, z), P_{t+1}(x, y', z)]$. Therefore, we get the following expression:

$$\begin{aligned} P_t(x, y, z) &= p(x, y, z_t) \\ &+ \beta \mathbb{E}_t \left[\left(\delta + (1 - \delta) \mathbf{1}\{P_{t+1}(x, y, z) < B_{t+1}(x)\} \right) B_{t+1}(x) \right. \\ &\quad \left. + (1 - \delta) \mathbf{1}\{P_{t+1}(x, y, z) \geq B_{t+1}(x)\} \left((1 - s\lambda_{w,t+1}) P_{t+1}(x, y, z) \right. \right. \\ &\quad \left. \left. + s\lambda_{w,t+1} \int \max\{P_{t+1}(x, y, z), W_{t+1}(x, y', z)\} \frac{v_{t+1}(y')}{V_{t+1}} dy' \right) \right] \quad (10) \end{aligned}$$

$$\begin{aligned} &= p(x, y, z_t) \\ &+ \beta \mathbb{E}_t \left[\left(1 - (1 - \delta) \mathbf{1}\{S_{t+1}(x, y, z) \geq 0\} \right) B_{t+1}(x) \right. \\ &\quad \left. + (1 - \delta) \mathbf{1}\{S_{t+1}(x, y, z) \geq 0\} P_{t+1}(x, y, z) \right]. \quad (11) \end{aligned}$$

Note that $P_t(x, y, z)$ is independent of the identity of the competing firm y' . This is because if the worker remains at firm y in period $t + 1$, the output $P_{t+1}(x, y, z)$ is not affected by the competing offer (only the share received by the worker changes). If the worker switches, the firm y receives 0, while the worker receives a value $W_{t+1}(x, y', z) = P_{t+1}(x, y, z)$.

Combining equations (8), (9), and (11), we get the following expression for the match

⁵the continuation value of a match post-separation is $B_{t+1}(x)$, as the firm receives a value of 0

⁶Abusing the notation slightly, we remove the time subscript on the aggregate shock z . It should be interpreted as follows: If the value function is for the period t , for example $P_t(x, y, z)$, then $z = z_t$. If the value function is for period $t + 1$, for example $P_{t+1}(x, y, z)$ then $z = z_{t+1}$. Note that a worker's type x and a firm's type y remain fixed across periods.

⁷At this point, firm y cannot raise its wage offer anymore, but firm y' can still raise its offer by an ϵ amount and poach the worker, so we break ties in favour of the firm which has a higher surplus with the worker.

surplus:

$$S_t(x, y, z) = p_t(x, y, z) - b_t(x) + \beta(1 - \delta)\mathbb{E}_t \max\{S_{t+1}(x, y, z), 0\} \quad (12)$$

Proposition 1 in [Lise and Robin \(2017\)](#) states that the match surplus depends on time only through the aggregate productivity shock, and it does not depend on the distributions of vacancies, unemployed workers, or worker-firm matches.

III.E Vacancy Posting

There is no fixed cost required for the creation of vacancies. The cost of posting $v_t(y)$ vacancies is increasing and convex:

$$c(v_t(y)) = \kappa_0 \frac{v_t(y)^{1+\kappa_1}}{1+\kappa_1}, \quad \kappa_0 > 0, \kappa_1 > 0. \quad (13)$$

Let $J_t(y)$ denote the expected value of filling a vacancy. If a vacancy does not lead to a viable match, the firm receives 0 value. If a match is formed, the value to the firm depends on the surplus generated from the match:

$$\begin{aligned} J_t(y) = & \int_{\mathcal{X}} \frac{u_t^+(x)}{L_t} \max\{S_t(x, y, z), 0\} dx \\ & + \int_{\mathcal{X}} \int_{\mathcal{Y}} \frac{sh_t^+(x, y')}{L_t} \max\{S_t(x, y, z) - S_t(x, y', z), 0\} dy' dx. \end{aligned} \quad (14)$$

The firm's vacancy-posting condition is

$$\kappa_0 v_t^{\kappa_1}(y) = \lambda_{f,t} J_t(y), \quad (15)$$

which gives us the following expression for the number of vacancies created by each firm type y :

$$v_t(y) = \left(\lambda_{f,t} \frac{J_t(y)}{\kappa_0} \right)^{\frac{1}{\kappa_1}}. \quad (16)$$

Substituting for the job filling rate and aggregating over firm types gives us the equilibrium tightness:

$$\theta_t = \left(\frac{1}{L_t} \int \left(\frac{\alpha_M J_t(y)}{\kappa_0} \right)^{\frac{1}{\kappa_1}} dy \right)^{\frac{\kappa_1}{\kappa_1 + 0.5}}. \quad (17)$$

III.F Worker Flows

During the separations stage, some employed workers will join the pool of unemployed either due to the exogenous job destruction shock δ , or because their match is not viable anymore ($S_t(x, y, z) < 0$). The resulting stocks of unemployed and employed workers is:

$$u_t^+(x) = u_t(x) + \int_{\mathcal{Y}} \left(\delta + (1 - \delta) \mathbf{1}\{S_t(x, y, z) < 0\} \right) h_t(x, y) dy \quad (18)$$

and

$$h_t^+(x, y) = (1 - \delta) \mathbf{1}\{S_t(x, y, z) \geq 0\} h_t(x, y). \quad (19)$$

During the matching and poaching stage, unemployed workers receive a job offer from a firm y with probability $\lambda_{w,t} \frac{v_t(y)}{V_t}$ and form a match with the firm if it is viable ($S_t(x, y, z) \geq 0$). The stock of employed workers at a firm come from three sources: (i) workers who were not poached by other firms, (ii) workers poached from another firm, and (iii) workers hired from unemployment. The resulting distribution of unemployed and employed workers at the end of period t (or at the beginning of period $t + 1$) is given by:

$$u_{t+1}(x) = u_t^+(x) - u_t^+(x) \lambda_{w,t} \int \frac{v_t(y)}{V_t} \mathbf{1}\{S_t(x, y, z) \geq 0\} dy \quad (20)$$

and

$$\begin{aligned} h_{t+1}(x, y) = & h_t^+(x, y) \left[1 - \int s \lambda_{w,t} \frac{v_t(y')}{V_t} \mathbf{1}\{S_t(x, y, z) < S_t(x, y', z)\} dy' \right] \\ & + \int h_t^+(x, y') s \lambda_{w,t} \frac{v_t(y)}{V_t} \mathbf{1}\{S_t(x, y, z) > S_t(x, y', z)\} dy' \\ & + u_t^+(x) \lambda_{w,t} \frac{v_t(y)}{V_t} \mathbf{1}\{S_t(x, y, z) \geq 0\}. \end{aligned} \quad (21)$$

III.G Aggregate Quantities

Aggregate output is

$$Y_t = \int \int p_t(x, y, z) h_{t+1}(x, y) dy dx \quad (22)$$

and aggregate consumption is

$$C_t = Y_t + \int b(x) u_{t+1}(x) dx - \int \kappa_0 \frac{v_t(y)^{1+\kappa_1}}{1+\kappa_1} dy. \quad (23)$$

III.H Equilibrium

Given an initial distribution (u_0, h_0) and aggregate shock Π_z , a dynamic equilibrium is a collection

$$\{B_t, P_t, S_t, J_t, v_t, u_t, h_t, Y_t, C_t, \theta_t\}_{t \geq 0} \quad (24)$$

such that:

1. B_t , P_t , and S_t satisfy the Bellman equations (8), (11), and (12);
2. J_t is consistent with expected gains from hiring and poaching, as in (14);
3. vacancy posting satisfies the firm's first-order condition (15);
4. market tightness θ_t is determined by the matching function and by search and vacancy-creation choices;

5. unemployment and employment distributions evolve according to the worker-flow equations and the state-transition kernels;
6. goods-market feasibility holds in every period;

III.I Wages

An interesting property of our model is that we do not need wages to determine labor market flows at any stage. All vacancy posting decisions and worker moves are decided based on the surplus of a match (Lise and Robin, 2017). However, we are also interested in tracking wages of workers, so we derive the expression for a worker's wages as described below.

Let σ_t be the share of surplus $S_t(x, y, z) = P_t(x, y, z) - B_t(x)$ that a worker x receives when matched with a firm y in period t . This share can depend on the history of past offers this worker has received. As in Lise and Postel-Vinay (2020), we assume that σ_t remains constant until the match dissolves or until an outside offer arrives and renegotiation happens:

$$\sigma_t(x, y, z) = \frac{W_t(x, y, z, \sigma) - B_t(x)}{P_t(x, y, z) - B_t(x)}. \quad (25)$$

Let $W_t(x, y, z, \sigma)$ denote the value a worker obtains from being employed, and let $w_t(x, y, z, \sigma)$ denote the wage that delivers this value. Then

$$\begin{aligned} W_t(x, y, z, \sigma) = & w_t(x, y, z, \sigma) \\ & + \beta \mathbb{E} \left[\left(\delta + (1 - \delta) \mathbf{1}\{S_{t+1}(x, y, z) < 0\} \right) B_{t+1}(x) \right] \\ & + \beta(1 - \delta) \mathbb{E} \left[\mathbf{1}\{S_{t+1}(x, y, z) \geq 0\} (1 - s\lambda_{w,t+1}(y)) W_{t+1}(x, y, z, \sigma) \right] \\ & + \beta(1 - \delta) \mathbb{E} \left[\mathbf{1}\{S_{t+1}(x, y, z) \geq 0\} s\lambda_{w,t+1}(y) \right. \\ & \quad \left. \times \min \{P_{t+1}(x, y, z), \max \{W_{t+1}(x, y, z, \sigma), P_{t+1}(x, y', z)\} \} \right]. \quad (26) \end{aligned}$$

The first term denotes current period wages. The second term considers the situation where the worker becomes unemployed next period.⁸ The third term refers to the case when the match is not destroyed and the worker does not receive a job offer from a competing firm. In this case, she receives the same share of surplus σ that she received in the previous period.

The final term provides the worker's value if an outside offer arrives. The actual value received by the worker depends on multiple cases: (i) If the outside offer is low enough, i.e. $W_{t+1}(x, y, z, \sigma) \geq P_{t+1}(x, y', z)$, where $P_{t+1}(x, y', z)$ is the maximum value that firm y' can offer, then the worker stays at her current firm and receives $W_{t+1}(x, y, z, \sigma)$, (ii) If the outside offer is large enough, i.e. $P_{t+1}(x, y', z) > P_{t+1}(x, y, z)$, then the worker switches to firm y' and receives a value of $P_{t+1}(x, y', z)$, and (iii) in the intermediate case when $P_{t+1}(x, y, z) \geq P_{t+1}(x, y', z) > W_{t+1}(x, y, z, \sigma)$, then the worker renegotiates her value up to $P_{t+1}(x, y', z)$ and stays at her current firm y .

⁸Note that even if the worker receives a job after becoming unemployed next period, her value remains $B_{t+1}(x)$ because the worker has zero bargaining power and only receives wage offers equal to her reservation value.

The expression for $w_t(x, y, z, \sigma)$ can then be written by rearranging Equation (26), using Equation (25) to obtain $W_{t+1}(x, y, z, \sigma)$ in terms of $P_{t+1}(x, y, z)$ and $B_{t+1}(x)$, and then substituting the expressions for $P_{t+1}(x, y, z)$ and $B_{t+1}(x)$ using Equations (11) and (8) respectively.

$$w_t(x, y, z, \sigma) = \sigma_t p_t(x, y, z) + (1 - \sigma_t) b_t(x) - \beta(1 - \delta) \mathbb{E} \left[s \lambda_{w,t+1} \mathbf{1}\{S_{t+1}(x, y, z) \geq 0\} \Omega_{t+1}(x, y, z, \sigma) \right], \quad (27)$$

where the renegotiation term is

$$\Omega_t(x, y, z, \sigma) = \int \left[\min \left\{ S_t(x, y, z), \max \left\{ S_t(x, y', z), \sigma_t S_t(x, y, z) \right\} \right\} - \sigma_t S_t(x, y, z) \right] \frac{v(y')}{V} dy'. \quad (28)$$

The expression inside the integral in $\Omega_t(x, y, z, \sigma)$ can be interpreted as follows:⁹

$$\min\{S, \max\{S', \sigma_t S\}\} - \sigma_t S = \begin{cases} 0, & S' \leq \sigma_t S \quad (\text{offer below current wage}) \\ S' - \sigma_t S, & \sigma_t S < S' \leq S \quad (\text{worker stays, renegotiates up}) \\ (1 - \sigma_t)S, & S' > S \quad (\text{worker leaves, firm loses its full share}) \end{cases}$$

In Equation (27), the first term denotes the share of surplus that the firm pays to the worker, over and above the worker's flow value of unemployment $b_t(x)$. The second term says that the firm anticipates outside offers in the future that might lead to wage increases and/or job switches, and it compensates for this possibility by lowering the wages offered in the current period.

IV Parametrization

To understand the impact of an increase in soft skill demand on labor market outcomes, we will first fit the model to key aggregate moments for the US labor market between 1951 and 2012. In this section, we document how well our model fits the data, and the parameter estimates we obtain.

We will target the means and standard deviations of labor market flows, such as the unemployment rate (U), and the transitions from unemployment to employment (UE), employment to unemployment (EU), and job-to-job (EE). We will also target moments related to the unemployment duration profile, i.e., the number of people unemployed for longer than 5 weeks (U^{5p}), longer than 15 weeks (U^{15p}), and longer than 27 weeks (U^{27p}). Aggregate time series data on most of the above moments are obtained from BLS for our time period in consideration, except for job-to-job transitions, for which we rely on CPS data between 1994 and 2012 (after applying the corrections proposed by Fujita et al. (2024)).

For labor demand side moments, we utilize data on the standard deviation of vacancies created (V), and the standard deviation and autocorrelation of firm value added (VA). Data

⁹It can also be written as $\min \left\{ (1 - \sigma_t)S(x, y, z), \max \left\{ 0, S(x, y', z) - \sigma_t S(x, y, z) \right\} \right\}$.

on vacancies is obtained from combining the BLS Help Wanted Index (HWI) for the period 1951-2006, and JOLTS for the period 2001-2012, where we project the HWI data on JOLTS data for the overlapping years to get a consistent time series. Time series data on firm value added is obtained through BLS. Additionally, we will use Compustat data to calculate the mean and standard deviation of the cross-sectional dispersion in firm value added (Bloom et al., 2018).

We also construct moments for the mean and standard deviation of labor market tightness (V/U), and for the correlations between value added and the multiple labor market variables (U , V , UE , EU). Finally, we also utilize data on the correlation between V and U , between UE and EE , and between VA and the cross-sectional dispersion in firm value added. The values for our data moments and the corresponding model moments are given in Table 4.

Table 4: Data Moments and Model Simulated Moments

Fitted moment	Data	Model	Fitted moment	Data	Model
$E[U]$	0.058	0.057	$sd[U]$	0.191	0.200
$E[U^{5p}]$	0.035	0.031	$sd[U^{5p}]$	0.281	0.311
$E[U^{15p}]$	0.018	0.017	$sd[U^{15p}]$	0.395	0.403
$E[U^{27p}]$	0.010	0.010	$sd[U^{27p}]$	0.478	0.420
$E[UE]$	0.421	0.474	$sd[UE]$	0.127	0.121
$E[EU]$	0.025	0.028	$sd[EU]$	0.100	0.094
$E[EE]$	0.025	0.025	$sd[EE]$	0.095	0.110
$E[V/U]$	0.634	0.756	$sd[V/U]$	0.381	0.303
$E[sd \text{ labor prod}]$	0.494	0.432	$sd[sd \text{ labor prod}]$	0.039	0.041
$sd[V]$	0.206	0.104	$corr[V, VA]$	0.721	0.999
$sd[VA]$	0.033	0.034	$corr[UE, VA]$	0.878	0.991
$autocorr[VA]$	0.931	0.994	$corr[EU, VA]$	-0.717	-0.960
$corr[V, U]$	-0.846	-0.988	$corr[UE, EE]$	0.695	0.987
$corr[U, VA]$	-0.860	-0.992	$corr[sd \text{ labor prod}, \Delta VA]$	-0.366	-0.275

Notes: U^{5p} , U^{15p} and U^{27p} are the fractions of the labor force unemployed more than 5, 15 and 27 weeks; UE , EU and EE are the monthly unemployment-to-employment, employment-to-unemployment and job-to-job transition rates; VA is aggregate value added, ΔVA its annual growth rate, and “sd labor prod” the cross-sectional standard deviation of log value added per job. The model is solved and simulated at a weekly frequency on a $21 \times 21 \times 51$ grid over (x, y, z) for 700 years, the first 100 of which are discarded, and the simulated series are then aggregated to the quarterly frequency (annual for the productivity-dispersion moments) exactly as in the data.

The model is solved at a weekly frequency, and simulated model moments are aggregated to the monthly or quarterly level to match the frequency at which the data moments are available.¹⁰ We set the discount rate to an annual rate of 5%, and matching elasticity ω is fixed at

¹⁰The reason why we choose a weekly model frequency is to reduce time aggregation bias. For example, if we decided to simulate the model at a monthly or lower frequency, we would be unable to capture the labor market transitions properly for a worker who loses her job at the beginning of the month and obtains another job at the end of the month. At a monthly frequency, this would be recorded as an EE transition although in reality it is an

Table 5: Parameter Estimates

		Estimate			Estimate
Matching	α	0.4957	Worker heterogeneity	β_1	2.1454
$\alpha L^\omega V^{1-\omega}$			Beta(β_1, β_2)	β_2	11.9311
Search intensity	s	0.0274			
Vacancy costs	c_0	0.0842	Value added	p_1	0.0025
$\frac{c_0}{1+c_1} v(y)^{1+c_1}$	$1 + c_1$	1.0288	$p(x, y, z) = z(p_1 + p_2x + p_3y$	p_2	2.0457
Exogenous separation	δ	0.0127	$+ p_4x^2 + p_5y^2 + p_6xy)$	p_3	-0.1393
Productivity shocks	σ	0.0657		p_4	8.0491
Gaussian copula (σ, ρ)	ρ	0.9997		p_5	-1.9195
				p_6	6.5988

Notes: This table shows the estimated parameter values from the simulated method of moments exercise. The discount rate is fixed at 5% per annum. The model is estimated at a weekly frequency. ω is fixed at 0.5.

0.5. The distribution of workers follows a beta distribution. The worker types x and firm types y are continuous on $[0,1]$, and they are each approximated using a linearly spaced grid having 21 points. The aggregate productivity shock z follows an AR(1) process, set up in the model using the discretized Rouwenhorst method, where $\text{mean}(z_t) = 1$.

To match the model to data moments, we use simulated method of moments (SMM). The SMM objective function is nonsmooth and has many local minima, so we use a mix of global search (using differential evolution) and local refinement (using Nelder-Mead) to obtain the set of parameter estimates that minimize the objective function globally. We simulate the model for 700 years (which includes an initial burn-in period of 100 years to minimize the effect of initial conditions). Table 5 shows the parameter estimates we obtain from the simulation.

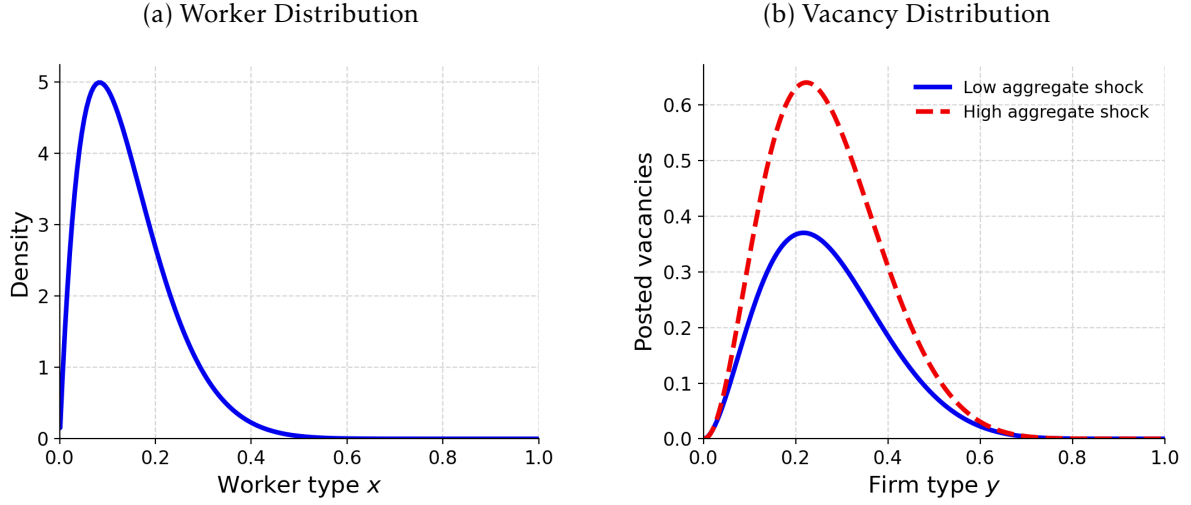
Even though none of the parameters have a clear one-to-one relationship with a moment, we can provide a rough description of which set of moments have a major influence on the identification of which parameters. The parameters for matching efficiency (α), relative search efficiency of employed workers (s), and exogenous separation rate (δ) govern movements in the labor market in the model, and they are identified primarily by the moments related to labor market flows. The parameters related to the aggregate shock process (ρ, σ) are identified by the standard deviation and autocorrelation of output.

The standard deviation of vacancies and the correlation of vacancies with output help us identify the vacancy cost function parameters (κ_0, κ_1). Worker heterogeneity (denoted by the beta distribution parameters β_1 and β_2) is identified by the moments for the unemployment duration profile. The idea is that the length of unemployment can indicate a worker's skill level, because high-skilled workers should find it relatively easier to receive job offers. Firm heterogeneity is identified by the cross-sectional variation in value added per job. The remaining standard deviation and correlation moments help us identify the production function parameters that govern complementarities between the firm and worker type.

We find an aggregate matching efficiency of 0.49 at the weekly frequency. Relative search

EU and UE transition.

Figure 3: Worker Types and Vacancy Creation



Notes: Panel (a) plots the distribution of worker types generated by the estimated parameters of the model ($Beta(\beta_1, \beta_2)$), where $\beta_1 = 2.14, \beta_2 = 11.93$. Panel (b) plots the distribution of vacancies created by firm types. The low (high) aggregate shock is equal to the 10th (90th) percentile in magnitude of the shock process.

efficiency of the employed workers is 0.027, which also accounts for the fact that not all employed workers search for jobs in all periods. The small, positive value of κ_1 indicates that the vacancy cost function is slightly convex. Jobs separate at an exogenous rate of 0.013 every week. The aggregate shock process is very persistent at the weekly frequency, with $\rho = 0.999$.

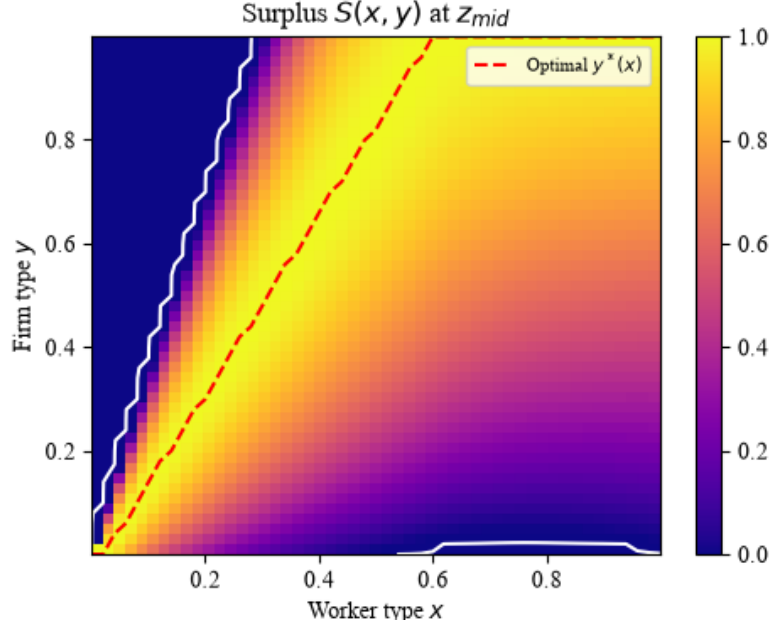
The worker distribution is shown in Figure 3a. The skewness of the distribution indicates that low-skilled workers are more common in this labor market. The vacancy distribution (as shown in Figure 3b) is similar in shape to the worker distribution. This is because the firm's vacancy creation decision depends not only on the skill levels of the workers, but also on their availability. This figure also shows that if the economy is in a productive state (high aggregate shock), more vacancies will be created since a higher surplus is generated in this state.

The production function parameters give us a key insight into the functioning of the model. p_1 is small, indicating that barely any output is produced by matches between the lowest firm and worker type. Keeping the firm type fixed, production is increasing in the worker type ($p_2, p_4 > 0$). Keeping the worker type fixed, output is non-monotonic in the firm type ($p_5 < 0$). For worker types $x < 0.6$, this implies a lower output produced if they match with firms of a relatively higher type.

Figure 4 shows the surplus generated by matches between different combinations of firms and worker types, calculated at the state when $z = z_{mid} = 1$. The white lines denote the boundary beyond which surplus becomes negative. The dashed red line gives the firm type y^* that maximizes the surplus produced for each worker type x . $p_5 < 0$ implies that y^* exists (even though it might be at the corner for some worker types), and $p_6 > 0$ implies that there is positive complementarity between the firm and worker types, leading to y^* being different for different worker types.

Figure 4 indicates how sorting takes place in this labor market. For low-type workers, matches with high-type firms are unviable. Intermediate worker types continue to switch em-

Figure 4: Match Surplus



Notes: This figure portrays the surplus generated by all possible worker-firm matches when there is no aggregate shock in the labor market ($z = z_{mid} = 1$). The white lines in the top left and bottom right areas delineate the boundary where matches create zero surplus. The dashed red line denotes the firm type $y^*(x, 1)$ which maximizes the surplus produced by a worker of type x .

employers until they reach as close to their maximum surplus generating firm type y^* as possible. For high-type workers, surplus is increasing in the firm type, so they are most likely to match with high-type firms.

V Counterfactual

In a counterfactual exercise, we attempt to analyze how much the youth unemployment rate rises in our model when we introduce a productivity shock similar in magnitude to the rise in soft skill demand during the period 2011-2024.

The current model does not include an age dimension, nor does it have two dimensions of skills to identify soft skills and technical skills separately. Therefore, we make the following assumptions. We will assume that $x \in [0, 1]$ denotes the level of soft skills a worker has, and this is fixed over time. Consistent with our claim that young workers do not have adequate soft skills, in the model, we will classify the 10% of workers at the lower end of the skills distribution as young workers.¹¹ Given the limitations of our current model, our current counterfactual exercise should be thought of as analyzing the impact on low-skilled workers, when the demand for skills increases.

We model the rise in skill demand as a productivity shock replicating skill-biased technical

¹¹In the CPS data, young workers make up 10% of the entire labor force.

Table 6: Counterfactual Estimates

	2011	2019	2024
Data	1.91	2.33	2.57
Model	4.62	4.98	5.21

Notes: This table shows the ratio between the unemployment rates of young v.s. older college graduates in different years. Young = 1 if age ≤ 25 , and 0 if age $\in [26, 55]$. A respondent is college educated if she reports obtaining a bachelor's or advanced degree. The data values are calculated from the CPS. The 2011 model value is the unemployment rate ratio in the baseline model without the increase in labor productivity. The 2019 (2024) model value is calculated by hitting the baseline model with a productivity shock equal in magnitude to the rise in soft skill demand between 2011 and 2019 (between 2011 and 2024).

change (SBTC). Consider the slightly modified production function:

$$p_t(x, y, z) = z_t(p_1 + p_2(mx) + p_3y + p_4(mx)^2 + p_5y^2 + p_6(mx)y)$$

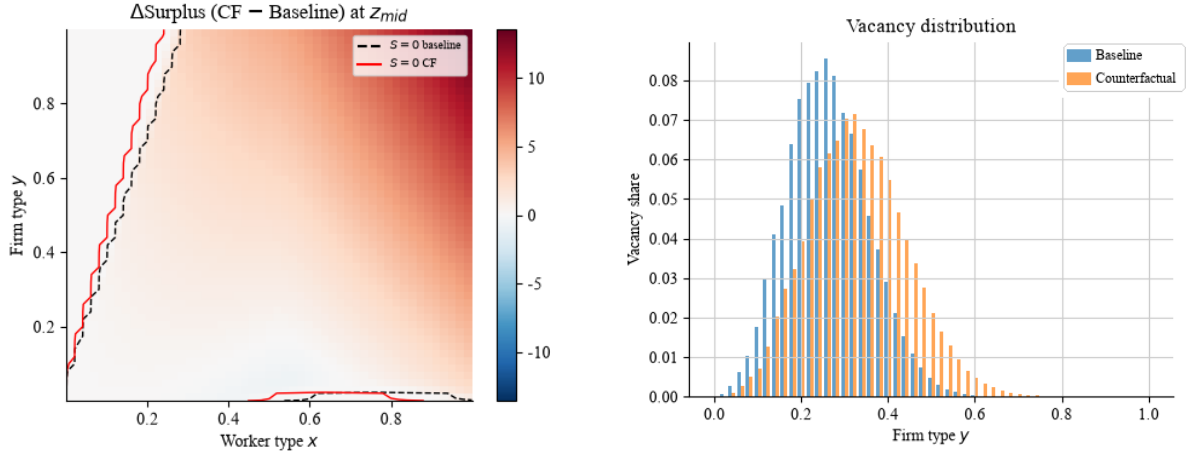
where m denotes the effective productivity of a worker of skill level x . In the baseline model, $m = 1 = m_{2011}$. SBTC is modeled as an increase in m equal to the increase in soft skill demand over 2011-2024. Therefore, we use $m_{2019} = 1.1$ and $m_{2024} = 1.16$ in the counterfactual. We refer to this technical change as skill-biased because higher-skilled workers experience a larger increase in their productivity:

$$\begin{aligned} \frac{\partial p_t(x, y, z)}{\partial m} &= z_t(p_2x + 2p_4x^2 + p_6xy) \\ \Rightarrow \frac{\partial}{\partial x} \left(\frac{\partial p_t(x, y, z)}{\partial m} \right) &= z_t(p_2 + 4p_4x + p_6y) \\ &> 0 \end{aligned}$$

Our object of interest is the ratio between the unemployment rate of young workers and the unemployment rate of old workers. We calculate this both in the data and in the model, for years 2011, 2019, and 2024 and report them in [Table 6](#). Comparing the values, we find that the unemployment rate ratio in the model increases 36% as much as the increase in the same object in the data. Therefore, we conclude that our model is able to explain 36% of the rise in youth unemployment even during the pre-COVID period 2011-2019.

Why does an SBTC shock lead to a relative rise in youth unemployment? This can be answered using [Figure 5](#). The SBTC shock makes all worker types more productive, so the set of viable matches (matches with a surplus ≥ 0) weakly expands for all workers. Firms also create more vacancies as more surplus is generated by the same worker type in a match. However, due to a larger increase in the productivity of higher-skilled workers, firms find it optimal to generate more vacancies for higher-type workers, leading to a shift in the vacancy distribution towards high-type workers. [Figure 6](#) shows the result. Even though the set of viable matches has increased for low type workers, there are fewer vacancies available for them. Their job finding rate increases in absolute terms, but it decreases in relative terms, and this leads to the relative increase in their unemployment rate.

Figure 5: Counterfactual: Change in Surplus and Vacancy Creation



Notes: Panel (a) shows the change in the set of viable matches after an increase in labor productivity, when $z = z_{mid} = 1$. The dashed black lines denotes the boundaries of the original set of viable matches in the baseline model. The solid red line shows the new boundaries after labor productivity increases. Panel (b) shows the the vacancy distributions before and after the increase in labor productivity.

VI Conclusion

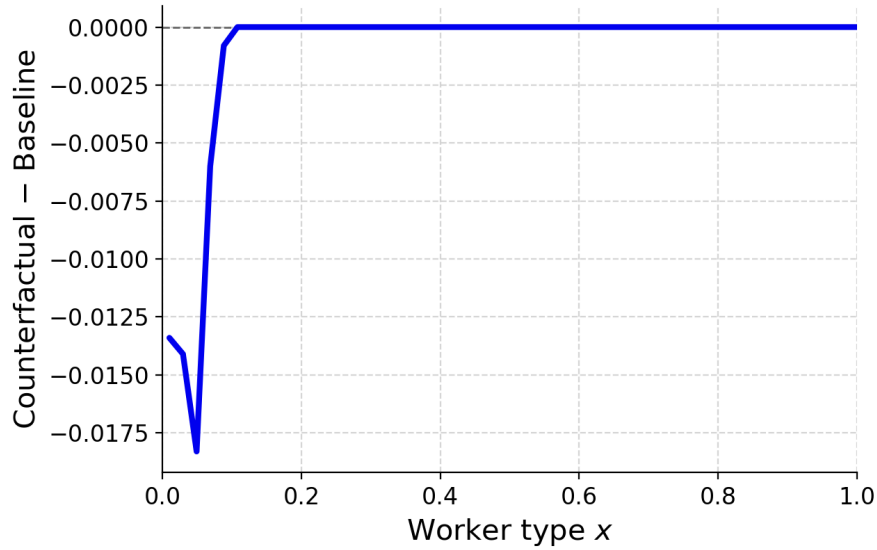
This paper set out to understand why young college graduates have fared increasingly poorly in the US labor market. We document two empirical facts. First, using near-universe data on online job postings, we show that employer demand for soft (noncognitive) skills has risen since 2011. Second, the gap between the wage of young and old workers is larger in jobs that are more soft-skill intensive. Combining both facts, we argue that as jobs become more soft-skill intensive over time, young workers would face increasingly worse labor market outcomes, as observed in the data as an increase in their unemployment rate.

We discuss that young workers fare worse in soft skill intensive jobs because soft skills are accumulated on the job, so young workers have lower levels of these skills when they enter the labor market. We build a general equilibrium, search-and-matching model, where we show that an increase in soft skill demand can account for 36% of the increase in the youth unemployment rate.

Thus far, our focus was on the unemployment rate. The rise in youth unemployment is analogous to an entry barrier to the labor market. It is possible that once young college graduates receive a job, they are able to acquire the relevant soft skills and do well in the labor market. However, similar to the related literature that finds long-run scarring effects on workers entering the labor market during a recession, it is possible that we find long-run negative effects of an increase in soft skill demand as well. Therefore, in future work, we also want to analyze the effect of soft skill demand on lifetime earnings.

Guvenen et al. (2022) show that the wages of young workers have remained stagnant since the 1990s. We also find a rising experience premium over the same time period. In our model setup, we would like to understand whether in a scenario where soft skill demand did not increase, or if young workers entered the labor market with the required amount of soft skills,

Figure 6: Fewer Viable Vacancies For Low-Skilled Workers



Notes: This figure shows the change in the share of viable vacancies available for each worker type $x \in [0, 1]$ in the counterfactual setting when labor productivity has increased. A vacancy from a firm type y is viable for a worker if the worker-firm match generates a positive amount of surplus.

their lifetime earnings would be higher or lower than what we see in the data.

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Appendix

A O*NET v.s. Lightcast

Our measure of an occupation’s skill requirements (or skill demand) is calculated using Lightcast data on job postings. As we described in [Section I](#), Lightcast is able to extract the skills that each job posting asks for. We classify these skills as soft or technical skills, and we define soft skill demand at the occupation $\times$ month level as the average number of soft skills per job posting within an occupation $\times$ month group (and similarly for technical skill demand). A drawback of using skills data from job postings is that we cannot observe the importance of each skill in a particular occupation. Instead, in our skill demand measure, all skills demanded in an occupation are weighted equally.

An alternative source of information on occupation skill requirements is the Occupational Information Network ([O*NET, 2026](#)). O*NET data consists of 35 skills and approximately 900 occupations, and it assigns an importance value between 1-5 and a level between 0-7 for each skill to each occupation. Importance signifies whether a skill is required in an occupation, and the level signifies the expertise in a skill that is needed in the occupation. For example, O*NET explains that the speaking skill is important for both lawyers and paralegals. However, lawyers are required to have a higher level in this skill than paralegals.¹² In this section, we will explore O*NET skills data to understand its suitability for our analysis.

O*NET updates its data on occupation skill requirements each year to capture any changes in the tasks involved in an occupation over time. We create binned scatterplots to compare how skill requirements have changed in occupations between 2011 (Version 16.0) and 2024 (Version 30.2). In [Figure A1](#) and [Figure A2](#), we see that for both the importance and level plots, the points lie close to the 45 degree line for both soft skills and technical skills, respectively. This would suggest that the O*NET occupational skill requirements remain relatively unchanged over the time period of our analysis.

Does this mean that skill demand is stable over our time period, and that our Lightcast measure of skill demand is incorrect? Here is why we think this is not the case. [Figure A3](#) shows binned scatterplots where we use our skill demand measure from Lightcast (average skills per posting in an occupation) instead of the O*NET skills requirement data. We observe that there is a near uniform increase in skill demand (both soft and technical) across occupations between 2011 and 2024, irrespective of the skill demand in an occupation in 2011.

O*NET data may be unable to capture this near-uniform increase in skill demand because of 2 potential reasons. Both reasons relate to the scales used to measure the importance and level of skill requirements in O*NET. First, given that the scales are fixed (1-5 for importance and 0-7 for level), an increasing trend in skill demand would eventually lead to all skill requirements reaching the upper bound of these scales if O*NET attempted to include this information in its measures. Therefore, we believe that although O*NET data is useful for comparing skill requirements across occupations at a particular time, it is not suitable for

¹²You can find the data dictionary for the latest O*NET Version at the time of writing here ([Version 30.2](#)). You can also find O*NET’s definition of soft skills [here](#).

Figure A1: O*NET Soft Skill Requirements (2011-2024)

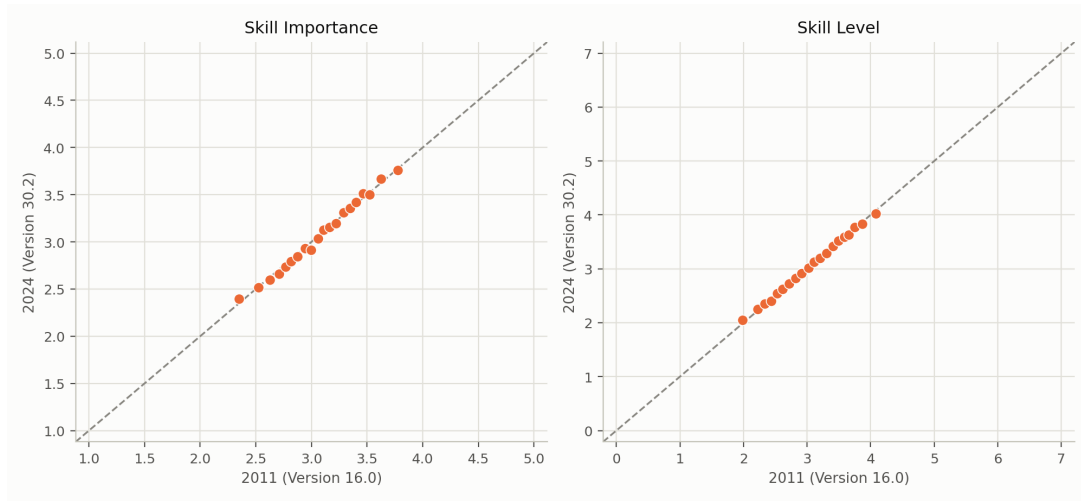
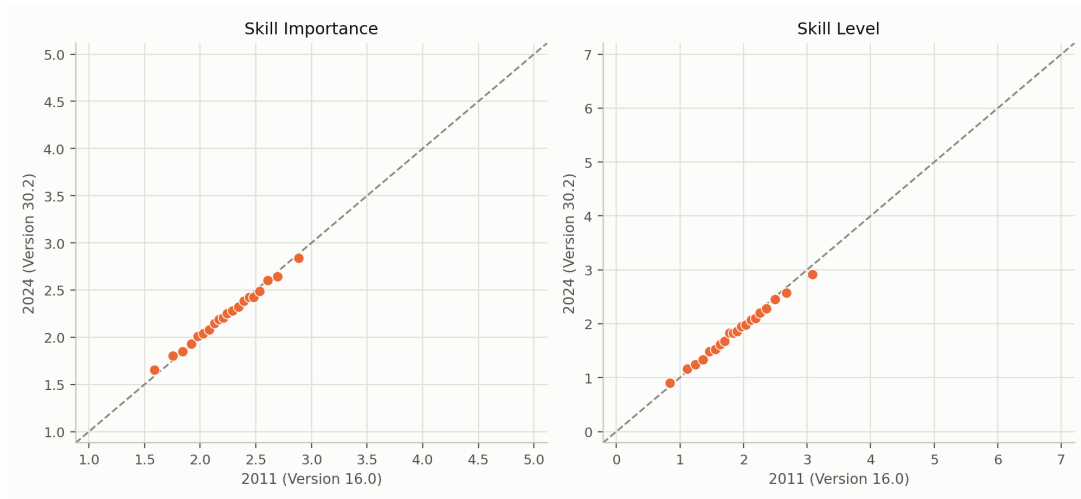


Figure A2: O*NET Technical Skill Requirements (2011-2024)



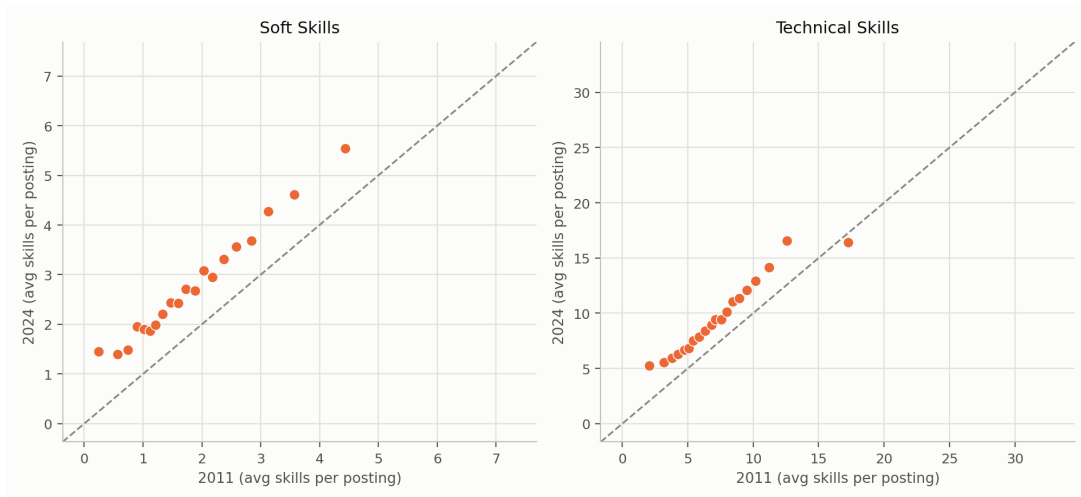
understanding the trends in these skill requirements over time. In short, O*NET should be used for studying relative skill demand across occupations, not absolute skill demand.

Finally, because all occupations experience a similar increase in skill demand (as observed in Figure A3), the relative skill requirements across these occupations also remains the same. If, for example, less soft-skill intensive occupations experienced a greater increase in soft skill demand, we would have seen a flatter line formed by the points in the binned scatterplot in Figure A1 relative to the 45 degree line.

To summarize, the O*NET and Lightcast data combined paint an interesting picture about the nature of the rise in skill demand. It suggests that the core tasks of an occupation remain unchanged over time (as captured by O*NET data). However, the rise in soft skill demand implies that all occupations now require an additional component, which requires workers to communicate and coordinate effectively with the people in their workplace.

We now show that O*NET and Lightcast capture similar information for occupations in Table A1. We first classify O*NET occupations based on whether they had above median

Figure A3: Lightcast Skill Requirements (2011-2024)



(High) or below median (Low) skill requirements in 2011. Our results are robust to using 2024 data as well. Then, we calculate the average number of skills required per job posting within these two groups using Lightcast data. Table A1 shows that for different skill dimensions (soft and technical) and skill measurement definitions (importance and level), the High group requires more skills per job posting than the Low group, and this difference is significant at the 95% confidence level. The last column shows that the rank correlation between the O*NET and Lightcast skill measures is positive as well.

To conclude, even though Lightcast data cannot measure the importance of different skills within an occupation like O*NET does, it still is able to differentiate between low and high-skill intensive occupations. Moreover, unlike O*NET, our Lightcast data is able to capture the rise in the demand for skills over time. Therefore, we choose to work with Lightcast data in our main analysis.

Table A1: O*NET vs. Lightcast: Construct Validity (2011)

O*NET Classification	Mean, High	Mean, Low	Difference	Rank Corr. (ρ)
Soft Importance	2.30	1.43	+0.87	0.43
Soft Level	2.28	1.45	+0.83	0.45
Technical Importance	8.34	6.31	+2.03	0.39
Technical Level	8.58	6.09	+2.49	0.47

High/Low = occupations above/below the 2011 O*NET median (Importance or Level, soft or technical, as indicated). Mean columns give the group's average Lightcast avg skills per posting in 2011. Rank Corr. is the Spearman correlation between the continuous O*NET score and the continuous Lightcast level, $N = 669$. All four High-vs-Low differences are significant at the 95% level.

Figure B1: Young Non-College Workers vs. Overall Unemployment Rate

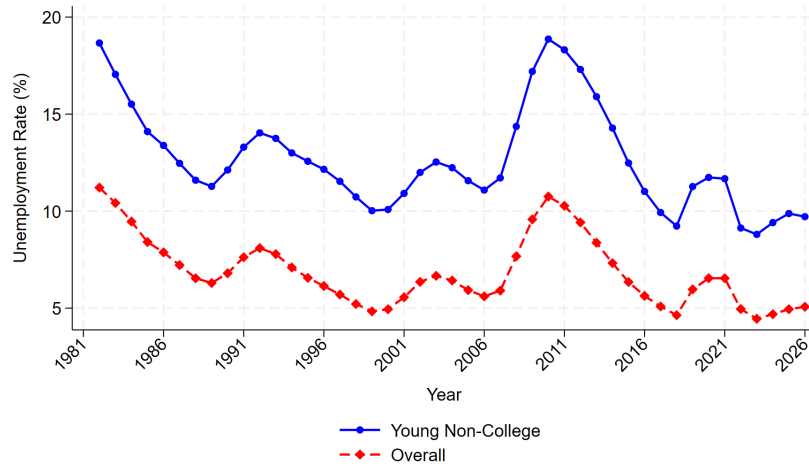
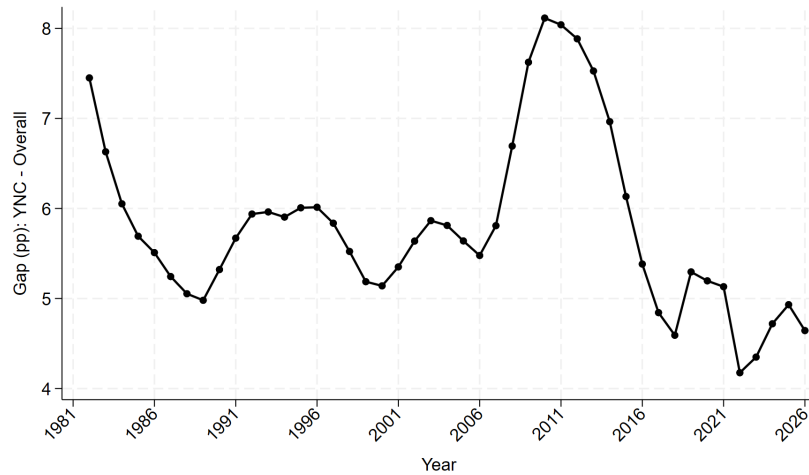


Figure B2: Gap Between The Young Non-College Workers vs. Overall Unemployment Rate



B Additional Empirical Analysis

B.1 Unemployment Rate of Non-College Graduates

In [Figure 1](#) in [Section II](#), we showed that the unemployment rate of college graduates has increased in recent times. On the other hand, [Figure B1](#) plots the unemployment rate of young non-college workers against the overall unemployment rate. We observe that the unemployment rate of this group since 2016 is lower than at any other time since 1981. [Figure B1](#) also shows that the gap between the two lines is smaller during the same time period. One reason why non college graduates do not experience the same rise in their unemployment rate that college graduates do, is because they have been in the labor market for a longer time. This means that they may have already accumulated the level of work experience and soft skills by the age of 25 that employers ask for.

B.2 Returns To Soft And Technical Skills For Young Workers

Table B1: Young $\times$ Soft Skills: Robustness

Specification	College Educated	Non-College Educated
Specification 1	−0.0300* (0.0142)	−0.0252** (0.0095)
Specification 2	−0.0329* (0.0158)	−0.0345** (0.0110)
Specification 3	−0.0325*** (0.0069)	−0.0173** (0.0066)
Specification 4	−0.0261 [†] (0.0141)	−0.0222** (0.0081)

Table B2: Young $\times$ Technical Skills: Robustness

Specification	College Educated	Non-College Educated
Specification 1	0.0093** (0.0033)	−0.0080** (0.0027)
Specification 2	0.0078* (0.0032)	−0.0092** (0.0030)
Specification 3	0.0000 (0.0018)	−0.0140*** (0.0029)
Specification 4	0.0085** (0.0033)	−0.0110*** (0.0026)

Figure B3: Young Non-College Workers vs. Overall Unemployment Rate

